MIL-C-28597B(YD)
9 July 1982
SUPERSEDING
MIL-C-28597A(YD)
10 December 1979

MILITARY SPECIFICATION

CLEANER, WATER BLAST, HIGH PRESSURE, TRAILER-MOUNTED, DED

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a high pressure, trailer mounted, diesel-engine-powered, water blast unit.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-V-173 - Varnish, Moisture-and Fungus-Resistant (for Treatment of Communications, Electronic, and Associated Equipment).
MIL-F-3541 - Fittings, Lubrication.
MIL-S-12651 - Sprayer, Packaging of.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. Not applicable.

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

* 3.1 Description. The water blast cleaner consists essentially of a diesel engine as the power unit; a positive displacement pump; reservoir water tank; air compressor; sand tank, hose and hose rack; controls, nozzles, safety valves; and necessary gages, tools, and auxiliary items. All components shall be securely mounted on a 2-wheel single or 4-wheel tandem trailer with wheel fenders, stabilizing jacks, heavy-duty tongue, and hitch for towing.

3.2 First article. When specified (see 6.2), the contractor shall furnish a water blast cleaner for first article inspection and approval (see 4.3 and 6.4).

3.3 Standard commercial product. The cleaner shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the cleaner being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.4 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.5 Interchangeability. All cleaners of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.6 Design. The unit shall be designed to permit easy accessibility for maintenance and service in the field. The design shall be such as to prevent conditions hazardous to personnel or deleterious to equipment.

* 3.7 Rated capacity. The cleaner shall produce a flow of not less than 4 gallons per minute (gpm) and not more than 10 gpm, and a controllable discharge pressure of at least 250 pounds per square inch (psi) up to a maximum of 10,000 psi.

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3.8 Construction. The cleaner s all include the items specified in 3.8.1 through 3.8.8.

3.8.1 Tank. The reserve water tank shall be a corrosion-resistant metal water tank of 30-gallon minimum capacity. The tank shall include an inlet float control valve that will function when connected to water supply systems having a pressure range of 25 to 125 psi. Suitable connections shall be provided between the tank and pump to insure an adequate water supply to the pump.

* 3.8.1.1 Filter. The cleaner shall be provided with a filter system of not less than 50 micron size having replaceable or cleanable type filters.

* 3.8.2 Pump. The pump shall deliver fresh or salt water at the specified volume and pressure. The pump liquid end shall be of corrosion-resistant metal construction. The pump shall be furnished with a pressure gage, and an adjustable stainless steel pressure relief valve. The pressure gage shall have a capacity range of 0 - 20,000 psi, and shall be graduated at intervals of 100 psi and numbered every 1,000 psi. The pressure-relief valve, factory-set at 11,500 psi shall shut off unit or bypass water when the pressure reaches 11,500 pounds.

* 3.8.3 Power unit. Power shall be supplied by a 3-, 4-, or 6-cylinder diesel engine of sufficient horsepower to provide continuous service with a variable load. The engine shall be complete with starter, generator, twin-disk clutch, remote controls, 12-volt battery with battery box and cables, high temperature, low oil cutoff controls, and fuel tank. The fuel tank shall have a minimum capacity of 20 gallons, be provided with a fuel level gage, and means for drainage. When specified (see 6.2), a tachometer, hourmeter, or a combination tachourmeter shall be furnished. When specified (see 6.2), the battery furnished in the unit shall be dry charged, without electrolyte, with sealed caps to prevent the intrusion of atmospheric moisture.

* 3.8.3.1 Air compressor. The air compressor furnished shall have the capacity and size characteristics to meet the requirements specified herein. The air compressor shall also have the capacity to meet requirements when the abrasive cleaning system is furnished.

3.8.4 Power takeoff. Power from the engine shall be transmitted to the pump by direct-, gear-, or V-belt drive. Direct- and gear-drives shall be totally enclosed. V-belts shall be enclosed by a removable cover to permit adjustment of belt tension.

* 3.8.5 Trailer assembly. The trailer assembly shall be the manufacturer's standard commercial item except as otherwise specified herein. The trailer shall be a 1- or 2-axle rigid frame with adjustable hitch or 4 wheel with steerable front axle, pneumatic-tired type complete with fenders, stabilizing jacks, lunette, hose rack, and tool box. Wheels shall be minimum 14-inch diameter.

3.8.6 Control device. The operator's control device shall be of the pistol-grip type having a shoulder stock, nozzles, and connectors. When the grip is released or the device is dropped, system pressure shall be automatically and instantaneously reduced to that of inlet pressure, making possible, safe, one-man operation.

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3.8.6.1 Nozzles. The nozzles shall be of the stainless steel, replaceable type which can be easily changed by the operator. Nozzles furnished shall include two each 0° straight, 15° flat fan, 40° flat fan, and 80° flat fan. When specified (see 6.2), tungsten carbide nozzles shall be provided.

* 3.8.7 Hose. Unless otherwise specified (see 6.2), two sections (approximately 100 feet) of 1/4- or 1/2-inch inside diameter wire braid hose having a minimum rated burst pressure of 20,000 psi shall be provided. Quick-disconnect couplings shall be furnished at both ends of each section of hose. When specified (see 6.2), a metal extension shall be provided between the hose and control device to protect the operator against water pressure should the hose rupture or become separated from the control.

3.8.7.1 Abrasive hose. When specified (see 6.2), one section of 3/4-inch inside diameter hose, 50-feet long shall be furnished to permit use of an abrasive during water blasting operations.

* 3.8.8 Accessories. When specified (see 6.2), a manufacturer's standard abrasive injector assembly shall be furnished, including sand tank.

3.9 Lifting attachments and tiedown devices. When specified (see 6.2), suitable lifting attachments and tiedown devices shall be furnished in accordance with the manufacturer's standard practice except as specified herein. The attachments and devices shall have eyes not less than 3 inches in inside diameter, and plates to withstand any handling conditions encountered. The tiedown devices shall be located at each corner of the trailer assembly. Adequate tiedown devices may also be used for lifting purposes. Each attachment or tiedown device shall be of sufficient capacity to carry the total weight. A copper or zinc-base-alloy or aluminum plate, securely fastened to the unit, shall indicate diagrammatically the lifting capacity of each attachment.

3.10 Lubrication. Unless otherwise specified (see 6.2), means for lubrication shall be provided in accordance with the manufacturer's standard practice. Parts requiring lubrication shall so located as to make the lubricating points easily visible and accessible.

3.10.1 Grease fittings. Grease fittings shall conform to MIL-F-3541 and shall be located in accessible protected positions. A bright red mark shall be painted at each lubrication point.

3.11 Fungus resistance. When specified (see 6.2), electrical components and circuit elements, including terminal and circuit connections, shall be coated with varnish conforming to MIL-V-173, except that:

a. Components and elements inherently inert to fungi or in hermetically sealed enclosures need not be coated.
b. Current-carrying contact surfaces, such as relay contact points, shall not be coated.

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3.12 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. The color of the finish coat shall be as specified (see 6.2). Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to insure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 1.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

3.13 Identification plate. The contracting officer will furnish to the Government inspector the required identification plates. The contractor will be required to stamp all necessary data in the blank spaces thereon and securely affix said plates in a conspicuous place on each unit, assembly and/or subassembly, and parts as directed by the Government inspector. Nonferrous screws, rivets, or bolts not less that 1/8 inch in diameter shall be used to affix the plates.

3.14 Instruction plates. When any special or important procedure is to be followed in operating and servicing the equipment, the unit shall be equipped with an instruction plate, suitably located, describing such procedure. Plates shall be of copper- or zinc-base-alloy, or aluminum, adequately secured.

3.15 Spare parts and maintenance tools. When specified (see 6.2), spare parts and maintenance tools shall be furnished. Special tools shall be kept to a minimum.

3.16 Workmanship.

3.16.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.16.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.16.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

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3.16 4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on one complete cleaner when a first article is required (see 3.2 and 6.4). This inspection shall include the examination of 4.3, the tests of 4.4, and, when specified (see 4.5 and 6.2), the preproduction pack inspection. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

* 4.2.2 Quality conformance inspection. Quality conformance inspection shall be performed as specified herein. This inspection shall include the examination of 4.3, the test of 4.4.1, and the packaging inspection of 4.5.

4.3 Examination. The cleaner shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

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4.4 Tests. The first article, when required, shall be tested as specified herein. Failure of any unit to pass any phase of any test shall be cause for rejection.

* 4.4.1 Performance tests. The cleaner shall be connected and operated in accordance with the manufacturer's operating instructions. With the control lever depressed, the cleaner shall be operated at the rated discharge of not less than 4 gpm at 10,000 psi for a period of 30 minutes. During this 30 minute period, the engine and pump shall be operated continuously, except that after each 5 minutes of operation, the control lever shall be released and the operation of the pressure reduction system observed for 1 minute. Pressure should be reduced to that of the inle pressure of the pump. Upon completion of this test, no component of the cle ner (engine, pump, hose connections, control mechanism, etc.) shall show evidence of failure, damage, or leakage.

* 4.4.2 Continuous operation test. With the control lever depressed, the cleaner shall be operated at rated discharge of not less than 4 gpm at 10,000 psi for a period of 48 hours. After each hour of operation, the control shall be released, and operation of the pressure reduction system observed for 1 minute. Upon completion of this test, there shall be no evidence of failure of, damage to, or leakage by any component of the cleaner (engine, pump, hose connections, control mechanisms, etc.,).

* 4.5 Packaging inspection. The inspection of the preservation, packing, and marking shall be in accordance with the requirements of section 4 of MIL-S-12651. The inspection shall consist of the quality conformance inspection; and, when specified (see 6.2), a preproduction pack shall be furnished for examination and test within the time frame required (see 6.2).

5. PACKAGING

5.1 Preservation, packing, and marking. Preservation, packing, and marking shall be in accordance with the requirements of MIL-S-12651 with the level of preservation and the level of packing as specified (see 6.2).

6. NOTES

6.1 Intended use. The water blast cleaners covered by this specification are intended for use in field service and base shop areas. The cleaners eject a high pressure jet stream of water to remove mud or clay, rust, scale, grease, poorly bonded paint, etc., from military vehicles, structures, and construction equipment. Cleaners are also used to remove marine growth and barnacles from ships and piers.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. When a first article is required for inspection and approval (see 3.2 and 6.4).
c. When a tachometer, hourmeter, or tachourhmeter is required (see 3.8.3).
d. When dry charged battery is required (see 3.8.3).
e. When tungsten carbide nozzles are required (see 3.8.6.1).
f. Hose sections, if different (see 3.8.7).

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g. When metal extension is required (see 3.8.7).
h. When hose for use with abrasive is required (see 3.8.7.1).
i. When abrasive injector assembly is required (see 3.8.8).
j. When lifting attachments and tiedown devices are required (see 3.9).
k. Means for lubrication, if different (see 3.10).
l. When fungus resistance is required (see 3.11).
m. Color of finish coat required (see 3.12).
n. When spare parts and maintenance tools are required (see 3.15).
o. When a preproduction pack is required and time frame required for submission (see 4.2.1 and 4.5).
p. Level of preservation and packaging and level of packing required (see 5.1).

* 6.3 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423 and invokes the provisions of paragraph 7-104.9(n) of the Defense Acquisition Regulations (DAR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data shall be delivered in accordance with the contract requirements.

6.4 First article. When a first article inspection is required, the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1). The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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